

SQL Notes – DDL, DML, DCL and TCL

1. DDL (Data Definition Language)

DDL is used to define and manage the structure of database objects such as:

- Tables
- Databases
- Schemas
- Views
- Constraints

DDL commands mainly deal with the structure of the database.

Most DDL commands are auto-committed.

Main DDL Commands

Command	Purpose
CREATE	Create database objects
ALTER	Modify existing objects
DROP	Delete objects permanently
TRUNCATE	Remove all rows from table
RENAME	Rename objects

CREATE

Used to create new database objects.

Syntax

```
CREATE TABLE table_name (  
    column1 datatype,  
    column2 datatype,
```

```
...  
);
```

Example

```
CREATE TABLE Student (  
    id INT,  
    name VARCHAR(50),  
    age INT,  
    marks INT  
);
```

Explanation

This creates a table named `Student` with four columns:

- id
- name
- age
- marks

ALTER

Used to modify an existing table.

You can:

- Add columns
- Remove columns
- Modify datatype
- Rename columns

Add Column

Syntax

```
ALTER TABLE table_name  
ADD column_name datatype;
```

Example

```
ALTER TABLE Student  
ADD email VARCHAR(100);
```

Modify Column

Syntax

```
ALTER TABLE table_name  
MODIFY column_name datatype;
```

Example

```
ALTER TABLE Student  
MODIFY name VARCHAR(100);
```

Drop Column

Syntax

```
ALTER TABLE table_name  
DROP COLUMN column_name;
```

Example

```
ALTER TABLE Student  
DROP COLUMN age;
```

DROP

Used to permanently delete database objects.

Syntax

```
DROP TABLE table_name;
```

Example

```
DROP TABLE Student;
```

Result

- Entire table removed
 - All records deleted
 - Structure deleted permanently
-

TRUNCATE

Used to remove all rows from a table while keeping the structure.

Syntax

```
TRUNCATE TABLE table_name;
```

Example

```
TRUNCATE TABLE Student;
```

Result

- Deletes all records
 - Keeps table structure
 - Faster than DELETE
-

RENAME

Used to rename a table.

Syntax

```
RENAME TABLE old_name TO new_name;
```

Example

```
RENAME TABLE Student TO Students;
```

Difference Between DROP and TRUNCATE

Feature	DROP	TRUNCATE
Deletes data	Yes	Yes
Deletes structure	Yes	No
WHERE allowed	No	No
Table remains	No	Yes
Faster	Yes	Very Fast

Real-Time Example of DDL

Suppose a college wants to create a student database.

```
CREATE TABLE Student (  
    roll_no INT,  
    name VARCHAR(50),  
    department VARCHAR(20)  
);
```

Later, email support is added.

```
ALTER TABLE Student  
ADD email VARCHAR(100);
```

To remove all student records:

```
TRUNCATE TABLE Student;
```

To remove the entire table:

```
DROP TABLE Student;
```

Important Points About DDL

- DDL changes database structure.
- DDL commands are usually auto-committed.
- Mainly used by database administrators.

2. DML (Data Manipulation Language)

DML is used to manipulate data stored inside tables.

It deals with:

- Inserting data
- Updating data
- Deleting data
- Retrieving data

Main DML Commands

Command	Purpose
INSERT	Add new records
UPDATE	Modify records
DELETE	Remove records
SELECT	Retrieve records

INSERT

Used to insert new rows into a table.

Syntax

```
INSERT INTO table_name  
VALUES(value1, value2, ...);
```

Example

```
INSERT INTO Student  
VALUES(1, 'Arun', 18, 95);
```

Insert Multiple Rows

```
INSERT INTO Student  
VALUES  
(2, 'Kavi', 19, 90),  
(3, 'Rahul', 18, 85);
```

Insert Specific Columns

```
INSERT INTO Student(id, name)  
VALUES(4, 'Vijay');
```

Remaining columns become NULL.

UPDATE

Used to modify existing records.

Syntax

```
UPDATE table_name  
SET column_name = value  
WHERE condition;
```

Example

```
UPDATE Student  
SET marks = 100  
WHERE id = 1;
```

Update Multiple Columns

```
UPDATE Student  
SET age = 20,  
    marks = 88  
WHERE id = 2;
```

Update All Rows

```
UPDATE Student  
SET marks = 50;
```

Without WHERE clause, all rows are updated.

DELETE

Used to remove records from a table.

Syntax

```
DELETE FROM table_name  
WHERE condition;
```

Delete Specific Row

```
DELETE FROM Student  
WHERE id = 3;
```

Delete All Rows

```
DELETE FROM Student;
```

- Removes all records
 - Table structure remains
-

SELECT

Used to retrieve data from tables.

Syntax

```
SELECT column_name  
FROM table_name;
```

Select All Data

```
SELECT * FROM Student;
```

Select Specific Columns

```
SELECT name, marks  
FROM Student;
```

Select with Condition

```
SELECT *  
FROM Student  
WHERE marks > 90;
```

Difference Between DELETE and TRUNCATE

Feature	DELETE	TRUNCATE
WHERE clause	Allowed	Not Allowed
Deletes selected rows	Yes	No
Removes all rows	Optional	Always
Faster	Slower	Faster
Structure remains	Yes	Yes

Real-Time Example of DML

```
CREATE TABLE Employee (  
    id INT,  
    name VARCHAR(50),  
    salary INT  
);
```

Insert records:

```
INSERT INTO Employee  
VALUES
```

```
(1, 'Asha', 40000),  
(2, 'Ravi', 35000);
```

Retrieve records:

```
SELECT * FROM Employee;
```

Update salary:

```
UPDATE Employee  
SET salary = 45000  
WHERE id = 1;
```

Delete employee:

```
DELETE FROM Employee  
WHERE id = 2;
```

Important Points About DML

- DML manipulates data inside tables.
- DML operations can usually be rolled back.
- WHERE clause is important in UPDATE and DELETE.

3. DCL (Data Control Language)

DCL is used to control access and permissions in a database.

It decides:

- Who can access the database
 - What operations users can perform
-

Main DCL Commands

Command	Purpose
GRANT	Give permissions
REVOKE	Remove permissions

GRANT

Used to provide privileges to users.

Syntax

```
GRANT privilege_name  
ON table_name  
TO user_name;
```

Example

```
GRANT SELECT  
ON Student  
TO myuser;
```

Explanation

The user `myuser` can now:

- Read data from Student table
- Use SELECT queries

Grant Multiple Permissions

```
GRANT SELECT, INSERT, UPDATE  
ON Student  
TO myuser;
```

REVOKE

Used to remove permissions from users.

Syntax

```
REVOKE privilege_name  
ON table_name  
FROM user_name;
```

Example

```
REVOKE INSERT  
ON Student  
FROM myuser;
```

Now user cannot insert records.

Common Privileges

Privilege	Meaning
SELECT	Read data
INSERT	Add records
UPDATE	Modify records
DELETE	Remove records
ALL PRIVILEGES	Full access

Real-Time Example of DCL

Suppose a company database has:

- Admin
- Employee

- Manager

Admin gives only read access to employees.

```
GRANT SELECT
ON Employee
TO emp_user;
```

Manager gets full access.

```
GRANT ALL PRIVILEGES
ON Employee
TO manager_user;
```

Important Points About DCL

- DCL manages security.
- Used by database administrators.
- Controls user permissions.

4. TCL (Transaction Control Language)

TCL is used to manage transactions in databases.

A transaction is a group of SQL operations treated as a single unit.

Why Transactions Are Needed

Example:

In a banking system:

- Amount deducted from Account A
- Amount added to Account B

If the system crashes in between, data becomes inconsistent.

Transactions solve this issue.

Main TCL Commands

Command	Purpose
COMMIT	Save changes permanently
ROLLBACK	Undo changes
SAVEPOINT	Create checkpoints

COMMIT

Used to permanently save transaction changes.

Syntax

```
COMMIT;
```

Example

```
UPDATE Account
SET balance = balance - 1000
WHERE id = 1;

UPDATE Account
SET balance = balance + 1000
WHERE id = 2;

COMMIT;
```

Result

Changes are permanently saved.

ROLLBACK

Used to undo changes before commit.

Syntax

```
ROLLBACK;
```

Example

```
UPDATE Account  
SET balance = balance - 500  
WHERE id = 1;  
  
ROLLBACK;
```

Result

Database returns to previous state.

SAVEPOINT

Used to create checkpoints inside transactions.

Syntax

```
SAVEPOINT savepoint_name;
```

Example

```
UPDATE Account  
SET balance = balance - 500  
WHERE id = 1;  
  
SAVEPOINT s1;  
  
UPDATE Account  
SET balance = balance + 500  
WHERE id = 2;
```



```
ROLLBACK TO s1;
```

Result

Second update is undone.

Transaction Flow

```
START TRANSACTION
  ↓
Execute DML commands
  ↓
COMMIT → Save permanently
or
ROLLBACK → Undo changes
```

ACID Properties of Transactions

Transactions follow ACID properties.

Property	Meaning
Atomicity	All or nothing
Consistency	Database remains valid
Isolation	Transactions do not interfere
Durability	Committed data remains permanently

Real-Time Example of TCL

Online shopping system:

1. Product stock reduced
2. Payment deducted
3. Order created
4. Invoice generated

If any step fails:

```
ROLLBACK;
```

Otherwise:

```
COMMIT;
```

Important Points About TCL

- TCL manages transactions.
- Used with DML commands.
- Ensures database consistency.
- Very important in banking and payment systems.

Final Summary Table

Category	Full Form	Purpose
DDL	Data Definition Language	Defines database structure
DML	Data Manipulation Language	Manipulates table data
DCL	Data Control Language	Controls user permissions
TCL	Transaction Control Language	Manages transactions

Complete Command Summary

Type	Commands
DDL	CREATE, ALTER, DROP, TRUNCATE, RENAME
DML	INSERT, UPDATE, DELETE, SELECT
DCL	GRANT, REVOKE
TCL	COMMIT, ROLLBACK, SAVEPOINT